

Bridging Biology and AI Active Inference for Biologically Plausible Decision Making Models

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the next talk is by aswin Paul bridging biology and AI active inference for biologically plausible decision making models this is a 15minute [Music] video hello everyone uh my name is ashin PA I recently finished my PhD from monach University in Australia and is currently working as a machine learning researcher in versus AI so today I'm here to talk about using active inference to model biologically plausible decision making uh and thank you active inference Institute for organizing this Symposium let us start uh by mentioning a recent Cutting Edge experiment that was developed by cortical labs in Melbourne so what they managed to do is they developed a chip a silicon chip and they cultured real biological neurons like mice cortical culture and human cortical culture and they managed to couple this chip with a computer game like the game of pong that you see on the screen right now so the experimental result is that given meaningful feedback these neural cultures actually demonstrate relative improvement over time in playing this game as you can see from this graph right like the MCC and SCC neurons actually demonstrate that they can actually improve in game uh play when compared to Baseline groups like the silico um agent or the control group groups so given such advancements in experiments from a theoretical perspective active inference is a great candidate to model such experiments and so we have emerging literature and a lot of evidence um suggesting this so when we actually try and do modeling there is a clear need of scalability in the active inference literature where we are used to um modeling very small environments with very low number of states uh and uh very low number of actions Etc so in order to actually um use active inference in such scenarios we have to scale up the existing active inference algorithms and in the first part of the talk I will be talking about some algorithms and techniques that we can use to scale up modeling decision making and active inference and in the second half I will talk about using these algorithms uh to actually model the particular experiment that we saw earlier so when we talk about um scaling up active inference or scaling up decision- making in active inference the

first technique I want to talk about is evaluating μ decision making backwards in time and not forward in time as we usually see in the active inference literature so when we Define the expected free energy μ and μ compute which action should be taken given a particular observation u_h rather than if evaluating expected free energy or Computing It Forward in time as we see in techniques like sophisticated inference we can actually use dynamic programming and model decision making backwards in time and this result is actually published in paper u_h in this particular paper if you are interested in this particular algorithm and a more General version of this algorithm was recently published as the inductive inference algorithm μ by μ our colleagues in versus Ai and μ in that paper we note that μ inductive inference algorithm scale up decision making drastically and can be applied μ to μ bigger generative models μ another technique that we can employ in this regard is to actually learn prior preferences and in order to do that we actually take μ inspiration from the optimal control literature u_h and this particular paper and this particular paper introduces an algorithm called said learning that μ improves μ scalability and μ efficiently computes μ the optimal action μ when compared to μ Cutting Edge algorithms like the Q learning algorithm and a particular U Matrix in this algorithm called the said Matrix μ is actually very similar to u_h the prior preferences in active inference and we can actually learn this set Matrix orders of magnitude more efficient than μ classic reinforcement learning algorithms and what we do in μ in this paper is that we show that we can actually learn prior preferences using a similar learning rule u_h and once we learn this prior preferences then decision making is really easy because we don't really have to plan a lot forward in time and we also demonstrate that learning is robust in this particular algorithm and μ talking about computational complexity both these methods one is DPF me method μ that computes μ expected free energy backwards in time and the other one that μ learns PR preferences are actually the classical active inference we see in the literature or the recent μ sophisticated inference and both these methods actually learn μ μ or scales linearly with time u_h against the exponential computational complexity of decision making that we see in the active inference literature μ so we tested these two algorithms u_h in terms of its scalability and adaptability in μ State spaces μ as high as 900 u_h and what we saw is that when we actually μ introduced stochasticity into the grid these algorithms are more adaptable u_h than μ Cutting Edge reinforcement learning algorithms like $din q$ and we demonstrate that these algorithms are scalable which gives us confidence is to go ahead and model μ experiments like this and when it comes to modeling neuronal Dynamics we already have existing active inference

algorithms uh that model neuron dynamics that is from um CBS ricken developed by professor isur and colleagues and in one of their Works they actually talk about um a similar experiment where um neural cultures are actually performing a perception task uh which is called B blind Source separation and in a more recent work they also talk about decision making in neural cultures uh and um develop a decision-making model uh that can be equated to neuronal Dynamics so going into details um we have this algorithm called counterfactual learning algorithm uh and uh they develop a version of the variation free energy uh that is term to term equivalent to a neural network cost function uh and this is for a particular class of neural network and more details can be found in this particular paper and there are two terms of interest uh in this algorithm one is a Time dependent risk parameter um and a Time dependent State action mapping that this particular agent um uses for decision making and because of um how it is developed it is biologically plausible and can be shown that every term is one-on-one related with the cost function that a particular class of neuronal networks actually minimize so in one of our recent works we um generalized this algorithm to have memory and we tested it in a similar um control task like the card P game that we see on the screen and we can actually draw comparison with the card Pole game and the pong game because both of them are reactive task uh where are where where the agent is supposed to actually react to the environment uh and take decisions quickly uh to balance this pole or to deflect the ball back um so we observe that um compared to planning based algorithms um like DQ and DPFE the counterfactual learning algorithms perform really well in this particular um task and gives us confidence to go ahead and model uh the biological um neurons or the or the kind of experiments that we saw earlier right and in addition to being able to model this uh we AI have um the advantage of explainability in these algorithms unlike um deep learning algorithms that are black boxes so if we actually perturb the environment say in a particular episode we can actually see that the risk term is shooting up in that ex episode and um this um particular term captures um changes in behavior of the counterfactual learning agent and we can actually try to explain how decisions are being taken at every time Point uh unlike blackbox approaches so given these um several algorithms that we have can we use them to actually model um the manifestation of n natural intelligence uh in experiments like dish brain right so we for that we actually develop uh an experimental based generative model that is based on this particular experiment so for example um this is the pong game environment that they use in the experiment um um prec l so for example um we have the x coordinate of the ball that being communicated to the dish brain chip uh through 38 distinct frequencies in the experiment um

and uh the y coordinate of the paddle and the ball being communicated through a different voltage levels in the experiment um so for example like we have three observation modalities uh in this experiment and we note that these are the number of states and the ways of communicating that to the agent and we um uh enumerate the state space of our generative model um using this experimental details uh and match our generative model to be exactly like um the experimental generative process and we also have like the active inference parameters um in our generative model like uh the transition dynamics of a pom DP uh and counterfactual learning based State action mapping the risk term and the prior preference distribution Etc that we discussed in the talk earlier and given that we have this experimental based generative model we can model decision making um of in silico active inference agent and hope to compare them with the experimental result to draw insights and so on right so first of all we uh perform simulations using the counterfactual learning agent uh and we can see that counterfactual learning agent um with memory Horizon um as low as two is actually uh performing at par with the HCC group in terms of performance so here CFL um T stands for um CFL memory Horizon T So as expected with more memory Horizon it performs um better uh and uh we can actually look at um which memory Horizon is performing similar to the experimental group and so on and in one of our recent works we note that the active inference are incredibly data efficient compared to deep learning methods and it is only the active inference based model that is counterfactual learning uh learning method that demonstrate data efficiency at par with the biological neurons when compared to deep reinforcement learning algorithms like a2c and po if you are familiar with the Deep reinforcement learning um literature um so we also compare the planning based methods that we mentioned earlier like the dynamic programming method so in this particular task or the pong game um planning is not really useful so with increasing Horizon of planning um we don't really see um Improvement um and um planning Horizon of five performs as good as HCC neurons and planning Beyond a point is not useful in this particular task because of the nature of the task and we also not um the explainability of our models for example if we look at the average risk um in the counterfactual learning method we see that the risk is actually only reducing um significantly uh with a memory Horizon of four and which and this can also be correlated with the performance Improvement that we saw um in the results graph where with memory Horizon 4 there was a significant jump in performance and these kind of explainable insights um that can be drawn from our models as a clear advantage over black Mo methods like um deep reinforcement learning uh and we we can also look at um uh the entropy of the CL

parameters and we see that the entropy is decreasing over time that is um the evidence in learning and parameters reveal um differences in behavior of different agents and that is a clear Advantage here and to summarize um we actually developed an experimental um informed generative model and simulated decision making in the particular um uh environment uh we tested different active inference based algorithms in this setting and uh we also uh noted the similarity of memory augmented counterfactual learning method um to the biological neurons and we are interested in investigating similarities um of the memory based CFL method more the biological neurons and also insights like this can Inspire more experiments where um aspects like memory can be studied well uh there are several um limitations to this work where we haven't conducted um a really um involved statistical analysis in comparing these agents we are right now qualitatively comparing them um with um average parameters uh but we would like to do more statistical analysis uh and we would also like to um Inspire experiments from our theoretical predictions in our future work so I thank um all my collaborators in this work and if you're interested in this particular um work and the algorithms that I discussed uh you can actually um get access to all the papers here and uh feel free to contact me if you have suggestions or more questions about this work so thank you for your time uh and um I look forward to meeting you all when when I have a chance next thank you so much